Code No: 117EE **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, September - 2021
LINUX PROGRAMMING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1.a) Write a shell script to compute sum of digits of a number.
b) List and explain various File handling utilities available in Linux. [7+8]

2.a) What is the need of linking? How to create soft links between file? Illustrate the system call used to create soft links between files.
b) Explain fcntl( ) functions with an example program. [8+7]

3.a) What is the need of getenv( ) and setenv( ) system calls?
b) Describe how exec( ) family of system calls set the environment for the newly initiated processes. [5+10]

4. With a neat syntax and example, explain various APIs support for Message queues. [15]

5. Write down the steps and explain establishing connection between a client and a server machine using various socket system calls using connection oriented protocol. [15]

6.a) Write an awk script illustrating the need of Associative arrays.
b) Give a detail note on shell meta characters and shell built-in variables. [7+8]

7. List and explain various file permission system calls with clear syntax and example program. [15]

8.a) What are reliable and unreliable signals?
b) Illustrate kill( ) and raise( ) system calls with an example program. [7+8]

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